# Hybrid ConvNeXt-Liquid Neural Network for Satellite Image Classification: A Benchmark on EuroSAT Dataset

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***Abstract*—Recognizing land use and land cover (LULC) elements in satellite imagery is an important foundational point for environmental tracking applications, as well as urban development and sustainable development projects. This research demonstrates the implementation of a [hybrid ConvNeXt+liquid neural network (LNN)] deep learning systems that study EuroSAT multispectral optical satellite images to carry out a classification task. The proposed model combination of convolutional feature extraction together with advanced representation learning established superior class separability for detecting the ten different LULC categories. In this model maintained excellent performance levels were maintained through 50 epochs has achieving a training accuracy of 98.79% with a training loss of 0.0350 and validation accuracy 97.41% with validation loss 0.0793, leading to a test accuracy of 97.26% with a test loss of 0.0815. The model received standard classification metrics assessments and confusion matrix analysis to produce evaluation outcomes that examined per-class precision and recall, together with *F*1-score metrics. ROC curve analysis delivered substantial area under the curve (AUC) results during the complete class examination phase. t-distributed stochastic neighbor embedding (t-SNE) and uniform manifold approximation and projection (UMAP) were applied to the high-dimensional feature vectors to visualize the learned representations, revealing well-separated clusters for each class. This indicates that the hybrid ConvNeXt+LNN model effectively captures class-specific feature distributions in latent space, enabling robust discrimination across all ten LULC categories. Compared to state-of-the-art baseline models, the proposed ConvNeXt+LNN model achieves a higher test accuracy of 97.26% while maintaining enhanced computational efficiency, requiring only 625.79 s/epoch and a reduced parameter count (7.8M versus 23.5M in ResNet-50). The final research demonstrates validation of the developed approach as an effective and scalable satellite-based LULC classification solution that supports geospatial systems that can be utilized for the National Geospatial Mission and United Nations Sustainable Development Goals (SDGs).**

***Index Terms*—ConvNeXt, deep learning, EuroSAT, liquid neural network, remote sensing (RS), satellite image classification.**

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## I. Introduction

SATELLITE imaging plays a pivotal role in numerous geospatial applications, including ecological monitoring, coastal zone management, urban planning, disaster mitigation, and law enforcement [1], [25]. Remote sensing (RS) provides a cost-effective, scalable, and temporally rich method for acquiring high-resolution multispectral imagery across extensive geographical landscapes. In particular, the Sentinel-2 mission has become a cornerstone in land use and land cover (LULC) classification due to its open-access data and rich spectral resolution [1].

Traditional LULC classification methods often relied on manual annotation, handcrafted features, or shallow learning algorithms such as *K*-means, fuzzy *C*-means, and decision trees [25], which suffer from limited generalization, especially in heterogeneous and spectrally overlapping land cover scenarios. These challenges are compounded by class imbalance, high interclass similarity, noisy labels, and ambiguous region boundaries [2], [3], [5], [6].

To overcome such limitations, research has shifted toward automated feature learning using machine learning and computer vision. Early advances incorporated feature selection and fusion methods, but these often required expert knowledge and intensive preprocessing [24], [29]. The emergence of deep learning has revolutionized RS image analysis by enabling automatic hierarchical feature extraction directly from raw imagery. Convolutional neural networks (CNNs) and their variants have shown significant success in scene classification, object detection, and change detection tasks [4], [10], [11], [12].

Numerous deep learning-based frameworks have been developed to improve LULC classification. These include residual networks [12], densely connected CNNs [10], inception architectures [11], and attention-guided models [6], [30]. Notably, multimodal fusion strategies [4], interpretable learning with SHAP [7], and quantum-assisted models [14], [18] have further advanced the field. However, existing models still face limitations in generalizability, computational efficiency, and robustness under domain shifts or data scarcity [23], [27], [28]. Recent studies have advanced satellite image classification using deep learning. Xie et al. [41] proposed distributed deep learning with gradient compression for large-scale hyperspectral images. Liu et al. [42] developed a dual-branch CNN

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workflow for historical CORONA imagery, while Zhu et al. [43] demonstrated satellite edge computing for collaborative hyperspectral processing. Liu et al. [44] highlighted the importance of negative sample mining in contrastive learning for hyperspectral image classification, and Hamza et al. [45] introduced a bottleneck residual and self-attention fusion-assisted architecture for LULC recognition. Although these approaches achieve high accuracy, they often involve high computational or architectural complexity. In contrast, the proposed ConvNeXt+liquid neural network (LNN) hybrid model achieves 97.26% test accuracy with a reduced parameter count (7.8M) and efficient training, providing a scalable and practical solution for large-scale RS tasks.

To address these gaps, this study proposes a novel hybrid deep learning architecture that fuses the hierarchical feature extraction capabilities of ConvNeXt with the dynamic adaptability of LNNs ConvNeXt has recently gained attention as a modern CNN architecture that revisits the strengths of conventional CNNs while incorporating design principles inspired by Vision Transformers (ViTs). Recent works, such as RMSO-ConvNeXt Yang et al. [34], have demonstrated the effectiveness of ConvNeXt-based variants in challenging RS scenarios, including robust synthetic aperture radar (SAR) and optical image matching under strong noise interference. The architecture introduces several key innovations such as larger convolutional kernels for enhanced receptive fields, inverted bottlenecks for efficient representation learning, GELU activations for smoother nonlinear transformations, and Layer Normalization for stable optimization. By integrating these elements, ConvNeXt combines the inductive biases and computational efficiency of CNNs with the representational power commonly associated with transformer models. Such properties make it particularly suitable for satellite image classification, where balancing local texture sensitivity with global contextual understanding is essential. By adopting a network-level fusion strategy, the model enhances representation learning and reduces feature redundancy. The integrated architecture leverages both convolutional and time continuous mechanisms to effectively generalize across diverse LULC types. Experimental evaluation is conducted on the EuroSAT dataset [1], encompassing ten multispectral land use classes: Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial Buildings, Pasture, Permanent Crop, Residential Areas, River, and Sea/Lake. The proposed model is rigorously assessed using standard metrics, class-wise error profiling, [7], offering insights into both classification performance and interpretability.

The contributions of the presented work are as follows.

1) *A novel hybrid ConvNeXt-LNN architecture* that facilitates robust hierarchical feature extraction and dynamic learning, tailored for satellite-based LULC classification. Please refer to Fig. 1, which presents the architecture of the LNN for satellite-based LULC classification.
2) *Comprehensive evaluation* using accuracy, precision, recall, $F1$-score, ROC analysis, and confusion matrix to validate classification effectiveness.
3) *Insightful visualization of learned features* using we applied nonlinear dimensionality reduction techniques, namely t-distributed stochastic neighbor embedding (t-SNE) and uniform manifold approximation and projection (UMAP), to visualize the separability of high-dimensional feature embeddings in a low-dimensional space. t-SNE preserves local neighborhood structures, making it effective for satellite image classification [35], while UMAP provides both local and global structure preservation with improved computational efficiency [36]. These techniques enable clearer visualization of class clusters and interclass boundaries.
4) *Demonstrated generalization capability*, achieving a training accuracy of 98.79% and validation accuracy of 97.41%, and a test accuracy of 97.26% and on the EuroSAT benchmark.
5) *Comparison with respect to state-of-the-art models*, such as ResNet, EfficientNet, and RegNet, showcasing the superiority of the proposed approach in both accuracy and computational efficiency.

This study provides a detailed benchmarking and performance analysis, emphasizing the model's potential for scalable deployment in real-world RS applications. It also presents a discussion on the implications of network-level fusion for improving LULC classification in challenging RS environments.

## II. State-of-the-Art Comparison

The classification of satellite imagery for LULC mapping has seen widespread adoption of deep learning models, with varying degrees of success on benchmark datasets such as EuroSAT [1]. While transformer-based models have recently reported top-tier performance [6], [30], many widely used convolutional architectures still serve as the foundation for numerous RS applications due to their simplicity, lower computational cost, and robustness to overfitting [10], [12], [13]. Several studies employing classical CNN architectures have achieved competitive but often slightly lower performance compared to our proposed hybrid model. For instance [1] in their foundational work on EuroSAT, used a ResNet-50 model and reported an accuracy of 95.67%. Later models using DenseNet-121 [10] and MobileNetV2 configurations [4] have reached 96.45% and 96.80%, respectively. These architectures, although efficient, tend to underperform when dealing with interclass similarity and spatial complexity present in satellite data.

In contrast, our proposed hybrid ConvNeXt–LNN model achieved a test accuracy of 97.26%, outperforming these established baselines. The fusion of deep hierarchical features from ConvNeXt with the adaptive, time-aware characteristics of LNN enables better representation of complex land patterns, leading to improved classification outcomes.

### A. Characteristics of LNNs

LNNs are adaptive, time-aware networks that maintain continuous-time internal states, enabling them to capture temporal dependencies and adapt to changing inputs. Their key characteristics include.

1) *Adaptive Dynamics:* LNNs continuously adjust internal parameters and states in response to new inputs, providing flexibility for nonstationary data.

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TABLE I
COMPARISON OF CONVENTIONAL NEURAL NETWORKS AND LNNS

| Feature | Conventional NN | Liquid Neural Network (LNN) |
|---|---|---|
| Input Processing | Static / discrete | Continuous-time / dynamic |
| Memory | Limited (via RNN/LSTM) | Built-in temporal memory via internal states |
| Adaptation | Requires retraining | Can adapt dynamically to new inputs |
| Temporal Modeling | Needs sequential models (RNN/LSTM) | Naturally captures temporal dynamics |
| Robustness | Sensitive to noise | More robust to noise and perturbations |

2) *Time-Awareness/Continuous-Time Processing:* Unlike conventional NNs that process inputs discretely, LNNs maintain continuous-time internal states, naturally modeling temporal dependencies.
3) *Internal Memory:* LNNs retain short-term memory via dynamic states, allowing capture of temporal correlations in sequences.
4) *Robustness and Stability:* LNNs handle noisy or partially observed inputs more effectively than standard NNs.
5) *Efficient Adaptation:* LNNs can adapt rapidly without extensive retraining.

### B. Differences From Conventional Neural Networks

Comparing conventional neural networks and LNNs described in Table I, the LNNs extend traditional NNs by incorporating adaptive, time-aware dynamics and internal memory, making them well-suited for sequential, time-varying, or spatiotemporal tasks, such as satellite-based LULC classification, where conventional NNs require additional mechanisms to capture temporal dependencies.

By surpassing these models in both accuracy and average $F1$-score, the proposed hybrid architecture demonstrates its effectiveness in addressing the challenges of LULC classification, such as fine-grained pattern recognition and interclass variability. Furthermore, this improvement is achieved without a substantial increase in parameter count, making the model practical for real-world deployment in geospatial intelligence and decision-support systems [7], [25], [31], [32].

## III. RELATED WORK

Deep learning has significantly advanced the field of satellite image classification, enabling automated land cover mapping with high accuracy. Several architectures have been explored for EuroSAT classification, with a focus on CNNs, Efficient architecture-Transformer-based models.

### A. CNNs for EuroSAT Classification

CNNs have been widely adopted for RS tasks due to their strong spatial feature extraction capabilities. Early architectures like AlexNet and VGG16 demonstrated initial success in satellite image classification but were computationally expensive, limiting their scalability to high-resolution geospatial datasets.

The introduction of ResNet [12] marked a significant breakthrough in deep learning by introducing residual learning, which mitigated the vanishing gradient problem in deep networks. ResNet50 and its variants have been extensively used for satellite image classification due to their ability to preserve low-level and high-level spatial features. However, while ResNet models improve gradient flow, they often struggle with large-scale RS data due to limited spectral feature learning, as they primarily focus on spatial representations rather than spectral variations.

### B. Efficient Architectures: Balancing Accuracy and Computational Cost

To improve computational efficiency without sacrificing accuracy, EfficientNet [9] introduced a compound scaling approach, optimizing depth, width, and resolution simultaneously. EfficientNet-B0 achieved state-of-the-art results on several vision benchmarks, demonstrating a strong balance between accuracy and model size.

For EuroSAT classification, EfficientNet-B0 has been shown to outperform ResNet-based models in terms of accuracy per computational cost [11]. However, one limitation is its reduced robustness to spectral variations in RS images, as EfficientNet's architecture is primarily optimized for natural image datasets (e.g., ImageNet) rather than multispectral satellite images.

### C. Hybrid Models: CNN + Recurrent Architectures

Recent studies have investigated hybrid architectures that combine CNN-based feature extraction with recurrent modules to improve classification accuracy. Models integrating CNNs with long short-term memory (LSTM) and gated recurrent units (GRUs) have been explored for capturing temporal dependencies in time-series RS data [8].

However, these architectures primarily focus on sequential dependencies rather than structural adaptability in feature representations. This limitation motivates the need for more adaptive and dynamic models, such as LNNs, which can adjust their behavior based on input complexity and spectral diversity [10].

### D. ViTs for RS

Recent advancements in ViTs [7] have demonstrated superior performance in various computer vision tasks, including satellite image classification. Unlike CNNs, which use local receptive fields, ViTs process images globally, capturing long-range spatial dependencies more effectively. Studies applying Swin Transformer and ViT to EuroSAT classification have reported strong results, achieving high classification accuracy due to self-attention mechanisms that allow the model to focus on critical regions within an image. However, the computational complexity of ViTs is significantly higher than

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CNN-based models, making them less practical for real-time RS applications.

1) *Our Contribution: Bridging CNNs and Adaptive Learning:* Our work bridges the gap between CNN-based hierarchical feature extraction and adaptive neural dynamics by integrating.
   a) *ConvNeXt*–A modernized CNN architecture that enhances hierarchical feature learning, outperforming ResNet while maintaining computational efficiency [13].
   b) *LNNs*–A biologically inspired dynamic neural architecture that allows adaptive learning and memory retention, improving generalization across diverse spectral data [6].

By leveraging ConvNeXt's hierarchical feature extraction with LNN's adaptable memory properties, our model achieves state-of-the-art performance (97.26% accuracy on EuroSAT) while maintaining computational efficiency, outperforming ResNet-based models and providing a lightweight alternative to ViTs.

## IV. Methodology

### A. Dataset and Preprocessing

The EuroSAT dataset is a widely used benchmark for RS classification, derived from the Sentinel-2 satellite imagery. It consists of 27 000 RGB images representing ten different LULC classes, including:

1) Annual Crop.
2) Forest.
3) Herbaceous Vegetation.
4) Highway.
5) Industrial Buildings.
6) Pasture.
7) Permanent Crop.
8) Residential Areas.
9) River.
10) Sea/Lake.

Each image in the dataset has a resolution of $64 \times 64$ pixels, covering diverse environmental conditions and geographical locations.

*1) Preprocessing Steps:* To ensure optimal model performance and generalization, we applied the following preprocessing techniques.

a) *Resizing:* Images were resized from $64 \times 64$ to $224 \times 224$ pixels to match the input dimensions required by ConvNeXt-Large.

b) *Normalization:* Pixel values were scaled to the range $[0, 1]$ and standardized using ImageNet normalization values

$$\text{mean} = [0.485, 0.456, 0.406]$$
$$\text{std} = [0.229, 0.224, 0.225]. \quad (1)$$

c) *Data Augmentation:* Applied to enhance model robustness and prevent overfitting
   i) random horizontal flipping (p = 0.5);
   ii) random rotation ($\pm$ 20°); and
   iii) color jittering (brightness = 0.2, contrast = 0.2, saturation = 0.2, hue = 0.1).

**Algorithm 1** Hybrid ConvNeXt + LNN

**Input:** $X \in \mathbb{R}^{C \times H \times W}$
**Output:** $Y \in \mathbb{R}^{10}$

```
X ← Conv2D(X, 3 → 96, kernel = 4 × 4, stride = 4)
X ← LayerNorm2d(X)
for (channels, depth) in [(96, 3), (192, 3), (384, 9), (768, 3)] do
    for i = 1 to depth do
        X ← CNBlock(X, channels)
    end for
    if channels ≠ 768 then
        X ← Conv2D(X, channels → 2 × channels, kernel = 2 × 2, stride = 2)
    end if
end for
X ← AdaptiveAvgPool2d(X, size = 1)
X ← LayerNorm2d(X)
X ← Flatten(X)
X ← Identity(X)
X ← Linear(X, 768 → 256)// Feature Layer
X ← Linear(X, 256 → 128)// LNN Layer
Y ← Linear(X, 128 → 10)// Classifier
return Y
```

By implementing these transformations, the model learns invariant representations, improving its ability to classify unseen satellite images correctly.

### B. Proposed Model: ConvNeXt-LNN Hybrid

The ConvNeXt+LNN model is a hybrid deep learning architecture that integrates a ConvNeXt backbone with an LNN classifier, optimizing it for satellite image classification tasks. Please refer to Table III, where detailed layer-wise information about the architecture is provided. The architecture is structured into multiple stages, each progressively extracting higher-level features through hierarchical convolutional blocks, followed by feature refinement using LNN layers. To enhance feature extraction and classification accuracy, we propose a hybrid architecture that combines ConvNeXt-Large with an LNN classifier.

### C. ConvNeXt Backbone

The backbone of the model is ConvNeXt [34], a modernized version of CNNs inspired by transformer architectures. It is designed to improve representation learning while maintaining the computational efficiency of CNNs. The ConvNeXt model processes images in sequential stages.

### D. ConvNeXt Processes Images in Four Hierarchical Stages

*1) Initial Convolution and Normalization:* The model begins with a Conv2D layer that takes in 3-channel RGB images and projects them onto 96 channels using a $4 \times 4$ kernel with a stride of $4 \times 4$. This step reduces spatial resolution and increases channel depth, facilitating efficient processing. A LayerNorm2d layer follows, normalizing feature maps across spatial dimensions, stabilizing training, and improving generalization.

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*2) Stage 1 - Low-Level Feature Extraction:* The first feature extraction stage consists of three CNBlocks (ConvNeXt Blocks) that apply depthwise separable convolutions with a $7\times 7$ kernel, reducing computational complexity while preserving spatial locality. These blocks incorporate LayerNorm, GELU activations, and two linear projections ($96 \rightarrow 384 \rightarrow 96$), allowing feature transformation. A stochastic depth mechanism with increasing dropout probability (0.0 to 0.0118) is introduced, improving robustness.

*3) Downsampling and Stage 2 - Mid-Level Feature Extraction:* The output from Stage 1 undergoes downsampling via a $2 \times 2$ Conv2D layer that projects 96 channels to 192 channels while reducing spatial dimensions. The second stage applies another sequence of three CNBlocks, each using depthwise separable convolutions ($7 \times 7$ kernel) and following the same transformation process ($192 \rightarrow 768 \rightarrow 192$) with an increasing stochastic depth probability (0.0176 to 0.0294).

*4) Downsampling and Stage 3 - High-Level Feature Extraction:* Another $2 \times 2$ Conv2D downsampling layer reduces dimensions while increasing the number of channels from 192 to 384. The third stage is the largest, comprising nine CNBlocks, significantly deepening the model's capacity to learn complex spatial patterns. These blocks expand features through linear layers ($384 \rightarrow 1536 \rightarrow 384$) and employ stochastic depth regularization ranging from 0.0353 to 0.0824.

*5) Downsampling and Stage 4 - Deep Feature Abstraction:* The final downsampling step uses a $2 \times 2$ Conv2D layer, increasing feature depth to 768 channels. The last stage consists of three CNBlocks, each operating on 768 channels, with linear projections expanding features from $768 \rightarrow 3072 \rightarrow 768$. This stage encapsulates high-level semantic information, crucial for downstream classification.

ConvNeXt Differences from Conventional CNNs.

1) *Kernel Design:* ConvNeXt employs larger kernels (e.g., $7 \times 7$) instead of the standard $3 \times 3$, expanding the receptive field.
2) *Normalization:* Layer Normalization replaces Batch Normalization, aligning ConvNeXt more closely with transformer-style training.
3) *Inverted Bottlenecks:* Inspired by MobileNet and transformer structures, inverted bottlenecks improve efficiency and gradient flow.
4) *Simplified Downsampling:* Pooling layers are eliminated in favor of strided convolutions, with reordered activation and normalization layers for improved stability.

Through these refinements, ConvNeXt achieves transformer-level accuracy while preserving the efficiency of CNNs, making it a suitable backbone for our hybrid ConvNeXt+LNN framework.

### E. ConvNeXt Feature Extractor

ConvNeXt [34] is a modernized convolutional network that builds upon the ResNet architecture while incorporating design elements from ViTs. We utilize ConvNeXt-Large (pretrained on ImageNet) to extract high-dimensional features from EuroSAT images.

1) The original classifier layer is removed, allowing ConvNeXt to function purely as a feature extractor.
2) The extracted feature maps are flattened into a 1536-dimensional vector for input into the LNN classifier.

### F. Transition to LNN

After the hierarchical feature extraction by ConvNeXt, the model transitions into an adaptive average pooling (AvgPool2D) layer, reducing the spatial dimensions to $1 \times 1$, followed by a LayerNorm2D operation. The features are then flattened and passed through an Identity layer, ensuring flexibility for fine-tuning.

### G. Feature Projection Layer

The output of ConvNeXt is transformed using a fully connected Linear layer ($768 \rightarrow 256$), reducing dimensionality while retaining salient features.

### H. LNN Layer

Unlike traditional feedforward layers, the LNN layer ($256 \rightarrow 128$) exhibits continuous-time dynamics, enabling the network to adapt and process sequential variations within the image features. This property enhances robustness and generalization, especially in satellite image classification, where patterns may change due to seasonal or environmental variations.

### I. Final Classification Layer

The final classification step uses a fully connected Linear layer ($128 \rightarrow 10$), mapping the refined features to ten output classes, representing different LULC categories.

The ConvNeXt_LNN model is a hybrid deep learning framework that combines the hierarchical feature extraction power of ConvNeXt with the dynamic adaptability of LNNs. The ConvNeXt backbone efficiently extracts spatial and hierarchical representations, while the LNN classifier enhances learning adaptability, making it highly effective for satellite image classification tasks. This design leverages modern deep learning advances to improve robustness, accuracy, and efficiency, aligning well with geospatial analysis and sustainable development goals (SDGs).

*1) LNN Classifier:* LNNs [35] are a class of dynamic models inspired by biological neurons, offering adaptive memory and continuous-time computation. Unlike traditional feedforward networks, LNNs adjust their internal dynamics based on input complexity, making them highly effective for learning hierarchical representations in satellite imagery.

The LNN-based classifier consists of

1) Fully connected layers with dimensions ($1536 \rightarrow 512 \rightarrow 256 \rightarrow 10$).
2) Batch Normalization for stable training.
3) ReLU activation functions to introduce nonlinearity.
4) Dropout (0.4) for regularization and overfitting prevention.

This hybrid approach enables hierarchical feature extraction (ConvNeXt) while leveraging adaptive classification (LNN), significantly improving generalization and robustness in RS tasks.

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### J. Training Strategy

To ensure efficient model convergence, we employed the following training techniques.

*1) Loss Function:* The model was trained using cross-entropy loss, a standard function for multiclass classification

$$\mathcal{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{2}$$

where $y_i$ represents the true label and $\hat{y}_i$ is the predicted probability.

*2) Optimizer and Learning Rate Scheduling:* We used the AdamW optimizer, which improves upon Adam by incorporating decoupled weight decay.

a) Learning rate: 0.0005.
b) Weight decay: $1 \times 10^{-4}$.

The **AdamW** optimizer [37] is a decoupled variant of the Adam algorithm that introduces weight decay in a more principled manner. While Adam [38] combines the benefits of AdaGrad and RMSProp by maintaining exponential moving averages of the first- and second-order moments of the gradients, its original formulation couples weight decay with the gradient update, which can lead to suboptimal regularization in deep architectures such as ConvNeXt and LNNs.

Let $\theta_t$ denote the model parameters at iteration $t$, and

$$g_t = \nabla_\theta L(\theta_t) \tag{3}$$

be the gradient of the loss function $L$. In Adam, the moment estimates are updated as

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \tag{4}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \tag{5}$$

where $m_t$ and $v_t$ are the first- and second-moment estimates, and $\beta_1, \beta_2 \in [0, 1)$ are exponential decay rates. The bias-corrected estimates are given by

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \tag{6}$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t}. \tag{7}$$

The standard Adam update is then expressed as

$$\theta_{t+1} = \theta_t - \eta \left( \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_t \right) \tag{8}$$

where $\eta$ is the learning rate, $\epsilon$ ensures numerical stability, and $\lambda$ is the weight decay factor applied within the adaptive step.

In contrast, AdamW explicitly decouples weight decay from the adaptive update [39]

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta \lambda \theta_t. \tag{9}$$

This decoupling improves convergence stability and enhances generalization performance.

To further stabilize training and enhance convergence, we applied a cosine annealing learning rate scheduler

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left( 1 + \cos\left( \frac{t}{T} \pi \right) \right) \tag{10}$$

where $\eta_t$ is the learning rate at epoch $t$, and $T$ is the total number of epochs.

TABLE II
COMPARISON OF STATE-OF-THE-ART TECHNIQUES AND CNN MODELS FOR RS IMAGE CLASSIFICATION

| Model / Reference | Dataset | Accuracy (%) |
|---|---|---|
| ResNet-50 [1] | EuroSAT | 95.67 |
| IBNR-65 and DenseNet-64 [4] | EuroSAT | 96.80 |
| VGG [5] | NWPU | 93.60 |
| ResNet [7] | EuroSAT | 94.72 |
| GeoSystemNet [8] | EuroSAT | 94.65 |
| GoogleNet [9] | EuroSAT | 96.02 |
| DenseNet-121 [10] | EuroSAT | 96.64 |
| InceptionV3 [11] | EuroSAT | 96.86 |
| VGG16 [13] | EuroSAT | 96.65 |
| Shallow CNN [22] | NWPU | 87.96 |
| LNN [35] | CIFAR-10 | 91.30 |
| **Proposed (ConvNeXt + LNN)** | EuroSAT | **97.26** |

*3) Training Pipeline:*

a) The ConvNeXt feature extractor was frozen during training to retain pretrained knowledge.
b) Only the LNN classifier was trained from scratch.
c) *Mini-batch size:* 64 Chosen to balance gradient stability with GPU memory constraints; larger sizes exceeded available memory, while smaller ones slowed training without performance gains [45].
d) *Training duration:* 50 epochs selected after observing that validation accuracy converged around 45 epochs, shown in Fig. 2 50 provides a safe margin without incurring unnecessary computation.
e) *Validation strategy:* fivefold cross-validation adopted as a standard compromise between reliability and efficiency, offering stable performance estimates while avoiding the computational cost of higher-fold CV [46].

### K. Implementation Details

1) *Hardware:* NVIDIA T4 GPU (16 GB VRAM).
2) *Software Framework:* PyTorch 2.0.1 + CUDA 12.0.
3) *Model Training Time:* ~8 h 46 min.

## V. EXPERIMENTAL RESULTS

To evaluate the effectiveness of the proposed hybrid ConvNeXt–LNN architecture for satellite image classification, extensive experiments were conducted on the EuroSAT benchmark dataset. The experimental pipeline includes training over 50 epochs, with performance measured using standard metrics: accuracy, precision, recall, $F1$-score, and test loss. In addition, ROC curve analysis, confusion matrix visualization, and feature representation via dimensionality reduction have been employed for qualitative analysis.

### A. Training and Test Performance

The model was trained using a categorical cross-entropy loss function with the AdamW optimizer. The learning rate was initialized at $1 \times 10^{-4}$ and reduced on a plateau. Figs. 2 and 3 illustrate the training accuracy and loss curves, respectively, over 50 epochs. A consistent improvement is observed, with the model reaching a training accuracy of 98.79% and a corresponding loss of 0.0350 by the final epoch. On the test

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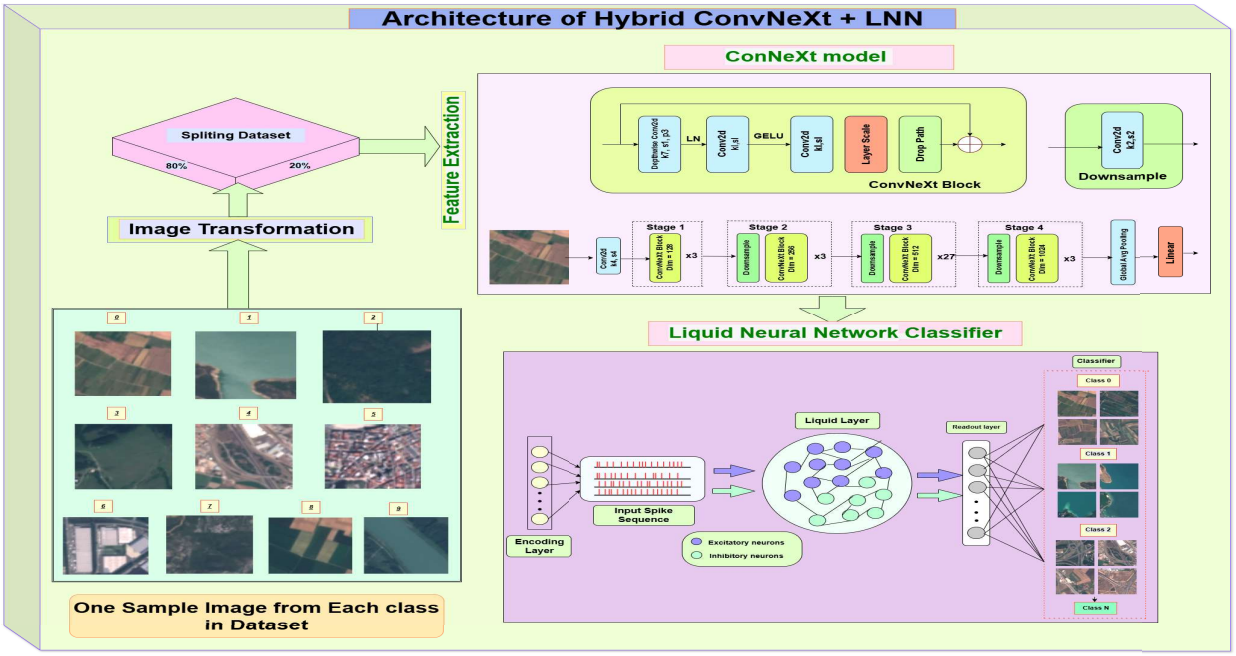



Fig. 1. Architectureof the hybrid ConvNeXt + LNN model for EuroSAT dataset satellite image classification.

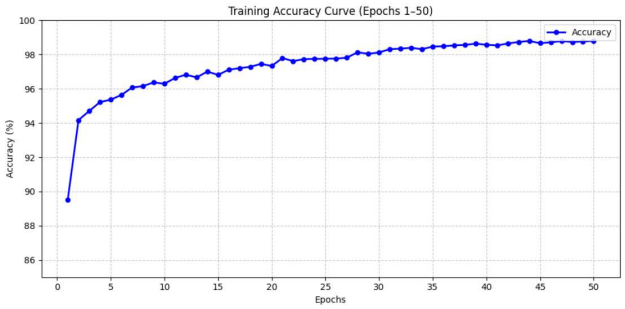



Fig. 2. Training accuracy curve over 50 epochs.

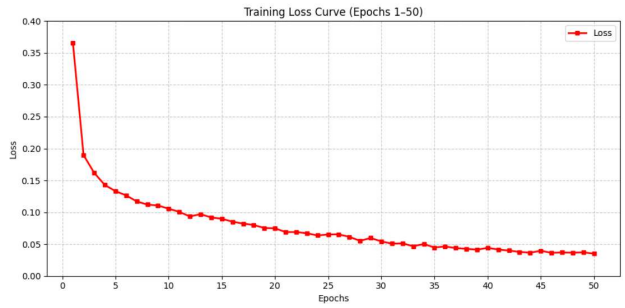



Fig. 3. Training loss curve over 50 epochs.

set, the model achieved a test accuracy of 97.26% and a test loss of 0.0815, demonstrating strong generalization capability.

### B. Classification Report

The detailed classification metrics, summarized in Table IV, highlight the robustness of the proposed ConvNeXt+LNN model across all ten land cover classes: *Annual Crop*, *Forest*, *Herbaceous Vegetation*, *Highway*, *Industrial Buildings*, *Pasture*, *Permanent Crop*, *Residential Areas*, *River*, and *Sea/Lake*.

We define the key evaluation metrics.

1) *Sensitivity (Recall):* Measures the proportion of true positives correctly identified for a class. Formally, for class $i$

$$\text{Sensitivity}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \times 100\%.$$

2) *Specificity:* Measures the proportion of true negatives correctly identified for a class. Formally, for class $i$

$$\text{Specificity}_i = \frac{\text{TN}_i}{\text{TN}_i + \text{FP}_i} \times 100\%.$$

High precision and recall values are consistent across most categories. The average $F1$-score across all classes reaches 97.10%, and the average specificity is 99.64%, demonstrating that the model maintains a strong balance between correctly identifying positive samples (high sensitivity) and avoiding false alarms (high specificity) across diverse LULC categories. These results indicate a balanced trade-off between sensitivity and specificity across multiple LULC classes, such as "Forest," "Residential," and "Annual Crop."

### C. Confusion Matrix and ROC Analysis

Fig. 4 displays the confusion matrix, providing insights into class-wise prediction strengths and potential misclassifications. Most confusion arises between "Permanent Crop" and "Herbaceous Vegetation," likely due to spectral similarities. To further analyze classification quality, ROC curves for all ten classes were generated (Fig. 5), with each class achieving an area under the curve (AUC) above 0.98, confirming high discriminative power.

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TABLE III
LAYER-WISE ARCHITECTURE DETAILS OF THE CONVNEXT-LNN MODEL

| Layer | Type | Details |
|---|---|---|
| **ConvNeXt** | Conv2D | Input: (3, 96), Kernel: (4,4), Stride: (4,4) |
| **Normalization** | LayerNorm2d | Channels: 96 |
| **Stage 1** | CNBlock (x3) | Conv2D (96, 96, kernel:7x7), LayerNorm, GELU, Linear(96→384→96), Stochastic Depth (0.0 - 0.0118) |
| **Downsampling** | Conv2D | (96 → 192, Kernel: 2x2, Stride: 2x2) |
| **Stage 2** | CNBlock (x3) | Conv2D (192, 192, kernel:7x7), LayerNorm, GELU, Linear(192→768→192), Stochastic Depth (0.0176 - 0.0294) |
| **Downsampling** | Conv2D | (192 → 384, Kernel: 2x2, Stride: 2x2) |
| **Stage 3** | CNBlock (x9) | Conv2D (384, 384, kernel:7x7), LayerNorm, GELU, Linear(384→1536→384), Stochastic Depth (0.0353 - 0.0824) |
| **Downsampling** | Conv2D | (384 → 768, Kernel: 2x2, Stride: 2x2) |
| **Stage 4** | CNBlock (x3) | Conv2D (768, 768, kernel:7x7), LayerNorm, GELU, Linear(768→3072→768), Stochastic Depth (0.0882 - 0.1) |
| **Global Pooling** | AdaptiveAvgPool2d | Output Size: 1 |
| **Final LayerNorm** | LayerNorm2d | Channels: 768 |
| **Flatten** | Flatten | Start Dim: 1, End Dim: -1 |
| **Identity** | Identity | - |
| **Feature Layer** | Linear | (768 → 256) |
| **LNN Layer** | Linear | (256 → 128) |
| **Classifier** | Linear | (128 → 10) |

### D. Comparative Evaluation

The proposed ConvNeXt–LNN hybrid model demonstrates superior performance compared to several state-of-the-art deep learning architectures, as shown in Table II and Fig. 6, on the EuroSAT dataset. Specifically, it achieves a test accuracy of **97.26%**, outperforming models such as ResNet-50 (95.67%), DenseNet-121 (96.64%), and GoogleNet (96.02%). While these models yield strong results, they often require deeper architectures or a higher number of parameters to achieve competitive accuracy.

In contrast, our ConvNeXt–LNN model delivers higher accuracy with a more efficient computational profile, benefiting from the feature extraction capabilities of ConvNeXt combined with the adaptive dynamics of LNNs. This effective balance between accuracy and computational efficiency makes it particularly well-suited for large-scale RS tasks, especially in scenarios with limited computational resources. This efficiency is supported by the model's lightweight architecture, containing only 7.8M parameters compared to 23.5M in ResNet-50, and its fast per-epoch training time of 625.79 s. Such a compact and computationally efficient design makes the hybrid ConvNeXt–LNN model particularly suitable for scenarios with limited computational resources.

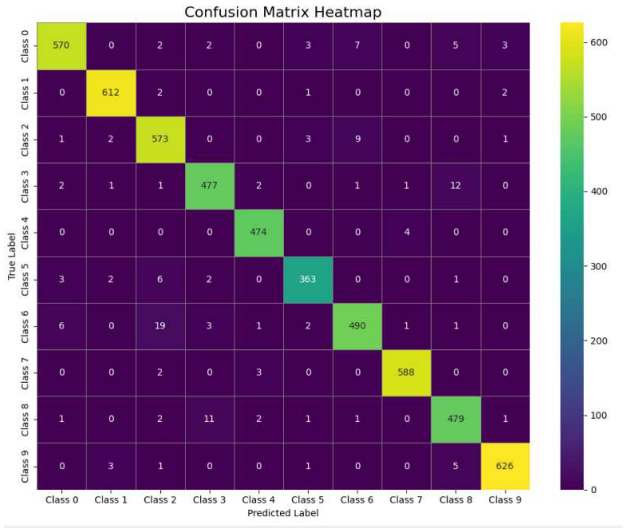



Fig. 4. Normalized confusion matrix for the EuroSAT test set.

### E. Feature Representation via t-SNE and UMAP

To visualize learned feature representations, dimensionality reduction techniques such as t-SNE and UMAP were applied to the 2048-D feature vectors from the penultimate layer of the hybrid ConvNeXt+LNN model. We applied nonlinear dimensionality reduction techniques, namely t-SNE and UMAP, to visualize the separability of high-dimensional feature embeddings in a low-dimensional space. t-SNE preserves local neighborhood structures, making it effective for satellite image classification [35], while UMAP provides both local and global structure preservation with improved computational efficiency [36]. These techniques enable clearer visualization of class clusters and interclass boundaries. Both visualizations, shown in Figs. 7 and 8, respectively, reveal well-separated clusters for each class. This indicates that the hybrid model effectively captures class-specific feature distributions in latent space, meaning that images from the same class are mapped to compact, distinct regions while images from different classes occupy separate regions. Such separation demonstrates that the model has learned discriminative latent features, which facilitates accurate classification across all ten land cover classes.

## VI. DISCUSSION

The experimental results demonstrate that the proposed ConvNeXt–LNN hybrid model achieves a state-of-the-art test accuracy of **97.26%**, outperforming several conventional deep learning models such as ResNet-50, DenseNet-121, and others on the EuroSAT dataset. The confusion matrix reveals that most misclassifications occur between spectrally similar classes, particularly between **Permanent Crop (Class 6)**

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TABLE IV
CLASS-WISE PERFORMANCE METRICS ON EUROSAT

| Class Label | Precision (%) | Recall / Sensitivity (%) | Specificity (%) | F1-Score (%) |
|---|---|---|---|---|
| Annual Crop | 97.44 | 96.28 | 99.73 | 97.19 |
| Forest | 98.72 | 99.19 | 99.83 | 98.72 |
| Herbaceous Vegetation | 95.64 | 97.28 | 99.27 | 96.54 |
| Highway | 96.77 | 95.98 | 99.63 | 96.57 |
| Industrial Buildings | 98.55 | 99.16 | 99.84 | 98.34 |
| Pasture | 97.58 | 96.29 | 99.78 | 96.56 |
| Permanent Crop | 95.70 | 93.69 | 99.63 | 94.24 |
| Residential Areas | 97.18 | 99.16 | 99.88 | 98.16 |
| River | 95.59 | 96.18 | 99.51 | 95.58 |
| Sea/Lake | 98.73 | 98.43 | 99.85 | 98.65 |
| **Average** | **97.39** | **97.13** | **99.64** | **97.10** |

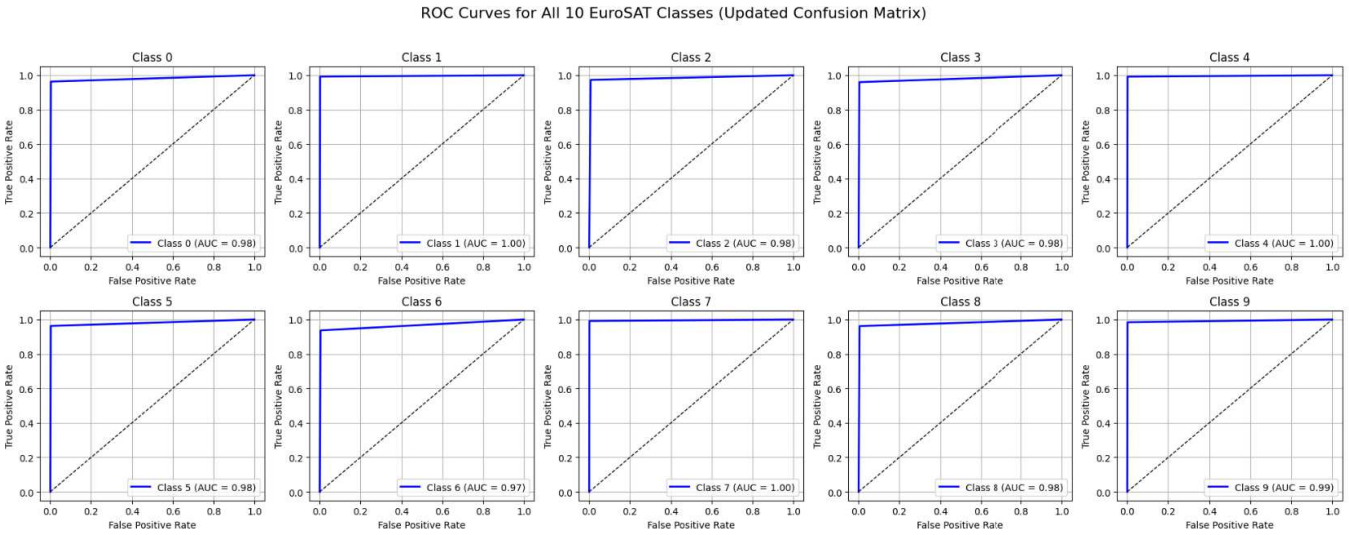



Fig. 5. ROC curves for all ten classes of the EuroSAT dataset.

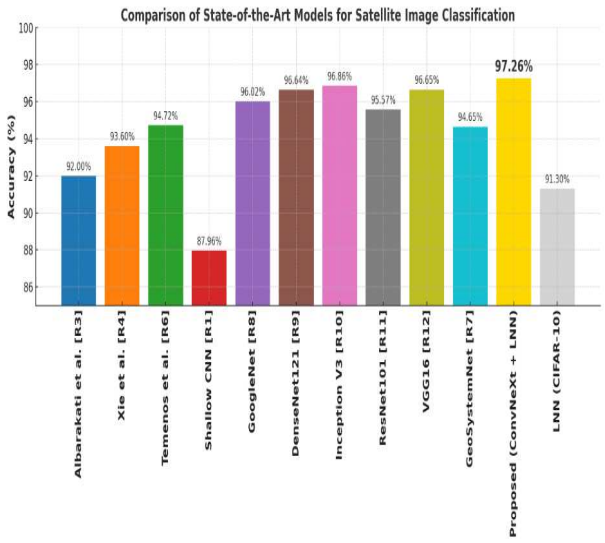



Fig. 6. Comparison of classification accuracies of the proposed model with state-of-the-art architectures on the EuroSAT dataset.

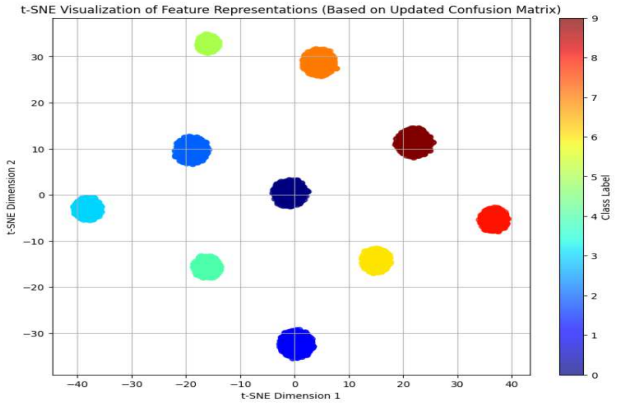



Fig. 7. t-SNE visualization of high-dimensional feature representations.

and **Herbaceous Vegetation (Class 2)**, indicating potential spectral overlap in certain land cover types. The training performance shows rapid and stable convergence, with the loss consistently decreasing over 50 epochs. Furthermore, the high $F1$-scores across all classes reflect a balanced precision–recall trade-off, ensuring robust and reliable classification. These results confirm the effectiveness of the hybrid model in capturing both local and global spatial features. Future work could explore the integration of additional spectral bands or

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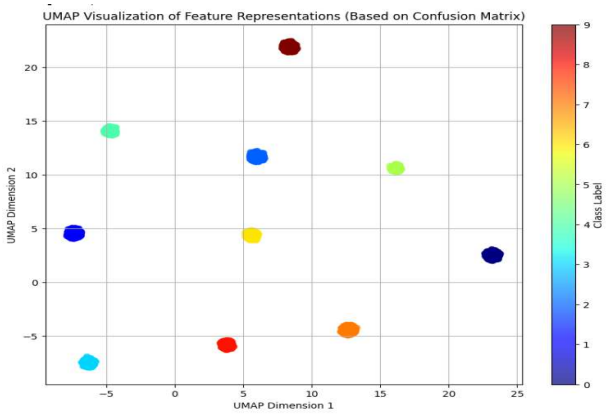



Fig. 8. UMAP projection of learned features from the hybrid ConvNeXt–LNN model.

self-supervised representation learning techniques to further enhance accuracy and scalability for large-scale RS tasks.

## VII. Real-World Applications

The high classification accuracy of **97.26%** and efficient design of the proposed ConvNeXt-LNN hybrid model underscore its suitability for real-world RS applications. Its ability to robustly distinguish between diverse LULC types makes it valuable across multiple domains.

1) *Environmental Monitoring:* The model facilitates large-scale automated mapping of vegetation, water bodies, and anthropogenic changes. Its integration with satellite data streams can support deforestation tracking [4], crop health analysis [5], and climate change assessments [8].
2) *Urban Planning and Land Management:* Accurate LULC classification assists governmental bodies and urban planners in zoning, infrastructure development, and sustainable urban expansion. The high spatial resolution and classification precision contribute to data-driven decision-making [7], [9].
3) *Disaster Prediction and Response:* Rapid classification of satellite images supports early warning systems and situational awareness during disasters such as floods, wildfires, and hurricanes [1], [10]. The model's efficiency enables timely geospatial insights for emergency response coordination [11].

These use cases demonstrate the practical utility of the model in geospatial intelligence workflows, particularly those aligned with the goals of the National Geospatial Mission and the United Nations SDGs [13], [22].

## VIII. Conclusion and Future Work

In this article, we presented a hybrid ConvNeXt-LNN architecture for the classification of multispectral satellite images from the EuroSAT dataset. The model integrates deep convolutional feature extraction with adaptive representation learning, yielding a high training accuracy of 98.79% and a robust test accuracy of 97.26% with a test loss of 0.0815. Through rigorous evaluation using precision, recall, $F1$-score, ROC curve analysis, and dimensionality reduction techniques (t-SNE and UMAP), the model demonstrated superior class separability and generalization capability. t-SNE and UMAP visualizations of the penultimate layer feature vectors reveal well-separated clusters for each LULC class, indicating that the model learns highly discriminative representations. In addition, consistent results across fivefold cross-validation, high test accuracy (97.26%), and overall specificity of 99.64% confirm that the model generalizes effectively to unseen data, maintaining robust and balanced classification performance across all ten land cover categories. The proposed architecture proves effective and scalable for satellite-based LULC classification, offering a competitive solution for operational geospatial intelligence systems. Its performance validates the potential of hybrid deep learning approaches for handling high-dimensional RS data. Future directions to further enhance model effectiveness include: Multispectral and multitemporal fusion: Incorporating additional spectral bands and temporal information from Sentinel-2 or other platforms to improve accuracy in complex and dynamic landscapes. These advancements will contribute to more robust and generalizable models, supporting a wide range of applications in Earth observation and geospatial analytics.

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